

Module -3

Mesh Analysis

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SENSE

Mesh Analysis

- A mesh is a loop which does not contain any other loops within it.

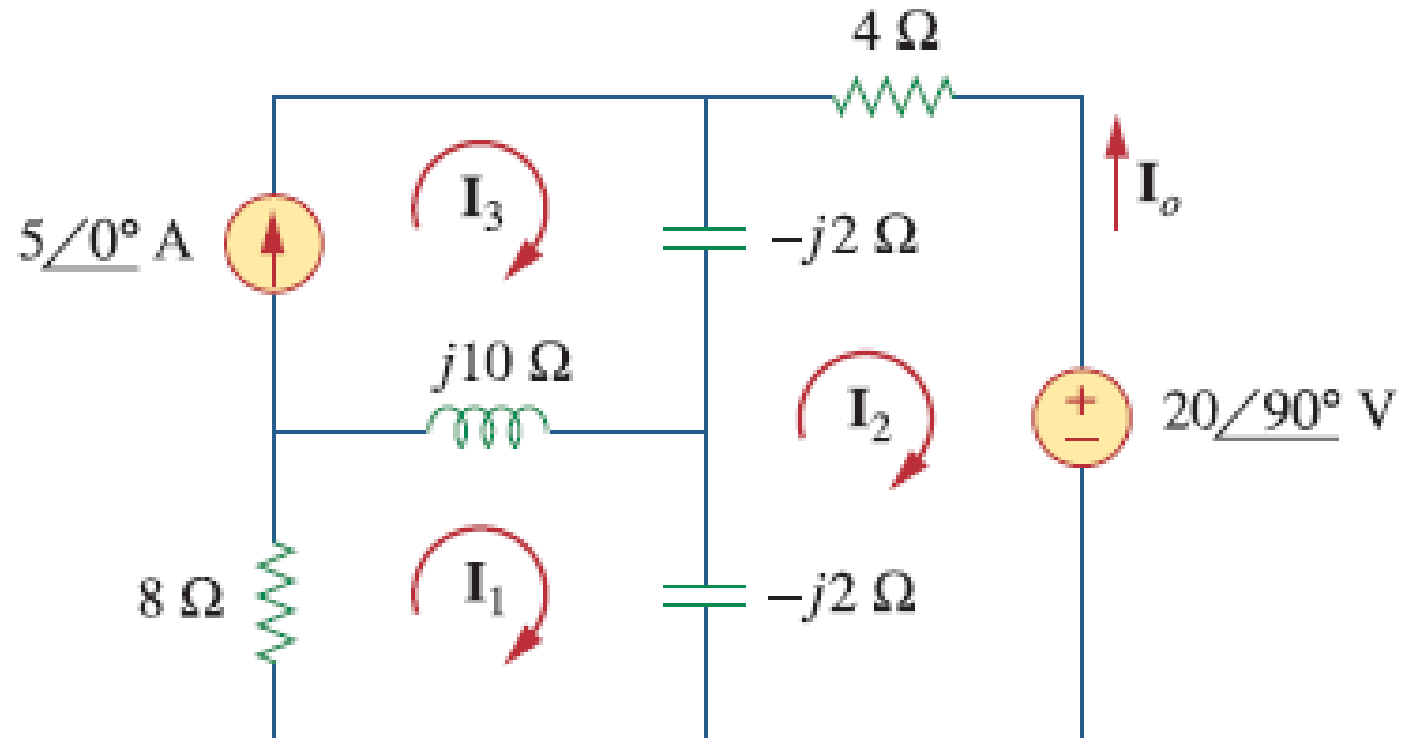
Steps to Determine Mesh Currents:

1. Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

Mesh Analysis

Numerical - 1

Determine current $\mathbf{I_o}$ in the circuit



Solution:

Applying KVL to mesh 1,

$$(8 + j10 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - j10\mathbf{I}_3 = 0 \longrightarrow (1)$$

For mesh 2,

$$(4 - j2 - j2)\mathbf{I}_2 - (-j2)\mathbf{I}_1 - (-j2)\mathbf{I}_3 + 20\angle 90^\circ = 0 \longrightarrow (2)$$

For mesh 3, $\mathbf{I}_3 = 5$.

Using $\mathbf{I}_3$ in eqn. (1) and (2)

$$(8 + j8)\mathbf{I}_1 + j2\mathbf{I}_2 = j50$$

$$j2\mathbf{I}_1 + (4 - j4)\mathbf{I}_2 = -j20 - j10$$

$$\begin{bmatrix} 8 + j8 & j2 \\ j2 & 4 - j4 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} j50 \\ -j30 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 8 + j8 & j2 \\ j2 & 4 - j4 \end{vmatrix} = 32(1 + j)(1 - j) + 4 = 68$$

$$\Delta_2 = \begin{vmatrix} 8 + j8 & j50 \\ j2 & -j30 \end{vmatrix} = 340 - j240 = 416.17 \angle -35.22^\circ$$

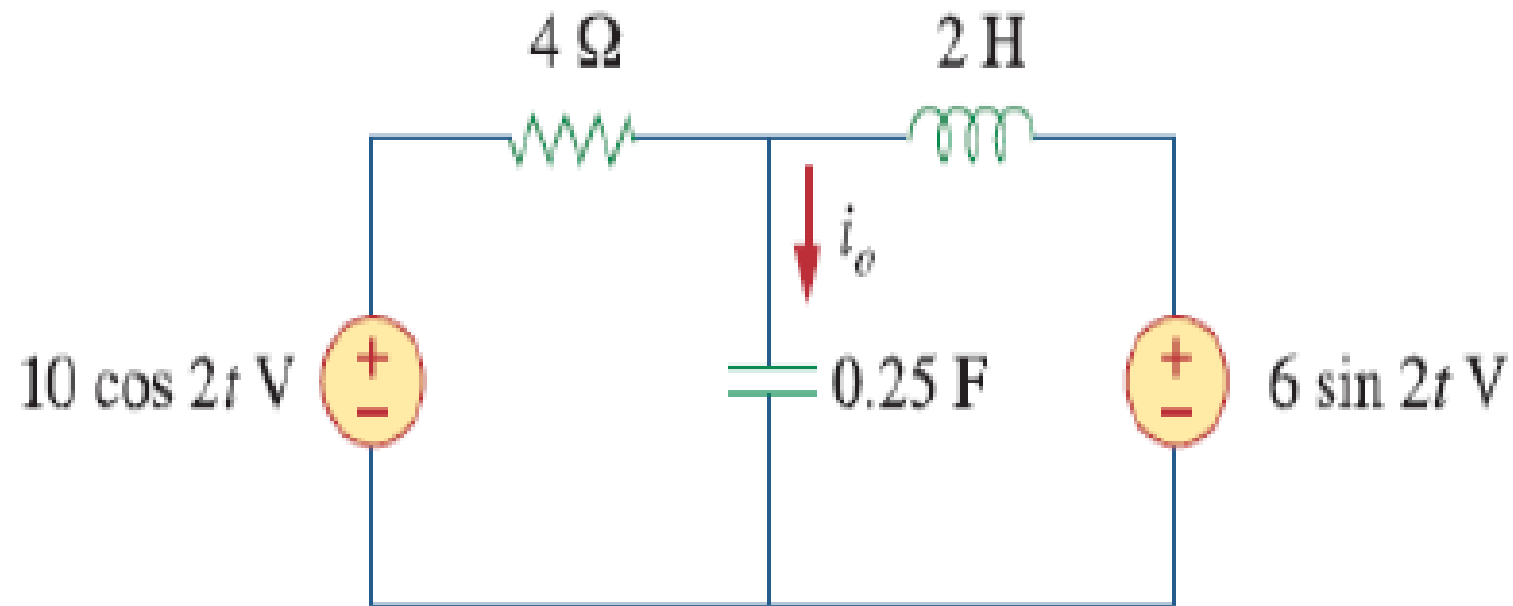
$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta} = \frac{416.17 \angle -35.22^\circ}{68} = 6.12 \angle -35.22^\circ \text{ A}$$

The desired current is

$$\mathbf{I}_o = -\mathbf{I}_2 = 6.12 \angle 144.78^\circ \text{ A}$$

Numerical – 2:

Find i_o of circuit in Fig. 1 using Mesh Analysis.



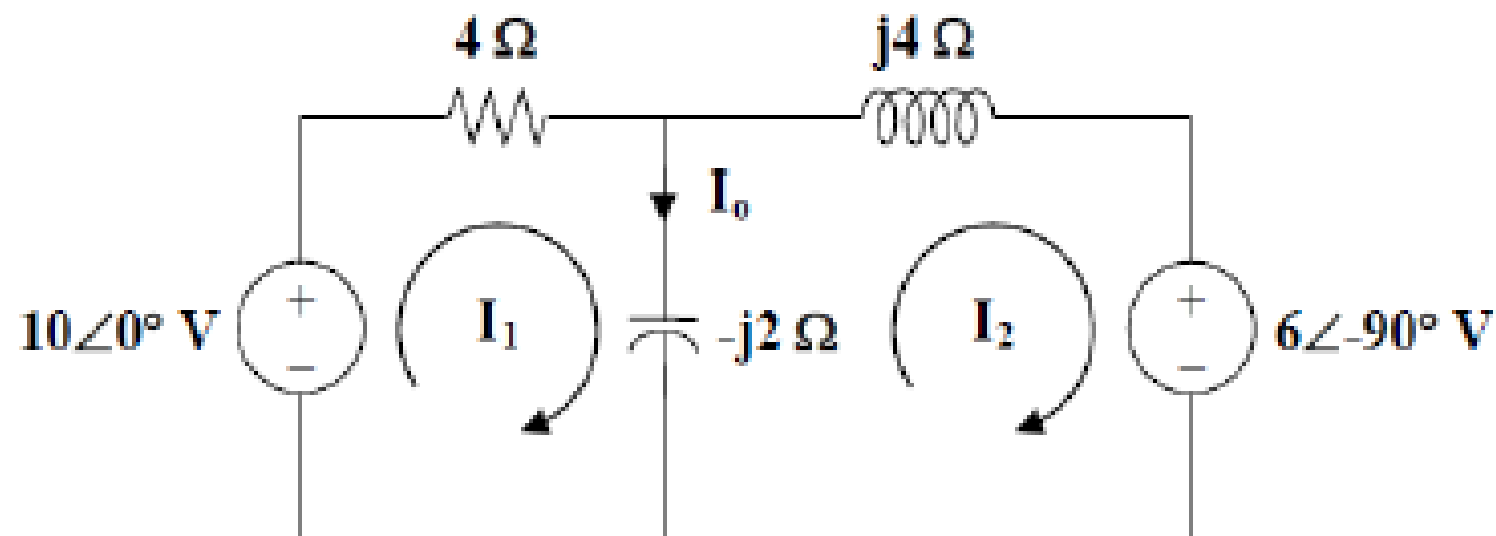
- Solution: $\omega = 2$

$$10 \cos(2t) \longrightarrow 10 \angle 0^\circ$$

$$6 \sin(2t) \longrightarrow 6 \angle -90^\circ = -j6$$

$$2 \text{ H} \longrightarrow j\omega L = j4$$

$$0.25 \text{ F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(2)(1/4)} = -j2$$



For loop 1,

$$\begin{aligned}-10 + (4 - j2) \mathbf{I}_1 + j2 \mathbf{I}_2 &= 0 \\ 5 &= (2 - j) \mathbf{I}_1 + j \mathbf{I}_2\end{aligned}\quad (1)$$

For loop 2,

$$\begin{aligned}j2 \mathbf{I}_1 + (j4 - j2) \mathbf{I}_2 + (-j6) &= 0 \\ \mathbf{I}_1 + \mathbf{I}_2 &= 3\end{aligned}\quad (2)$$

In matrix form (1) and (2) become

$$\begin{bmatrix} 2-j & j \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\Delta = 2(1-j), \quad \Delta_1 = 5-j3, \quad \Delta_2 = 1-j3$$

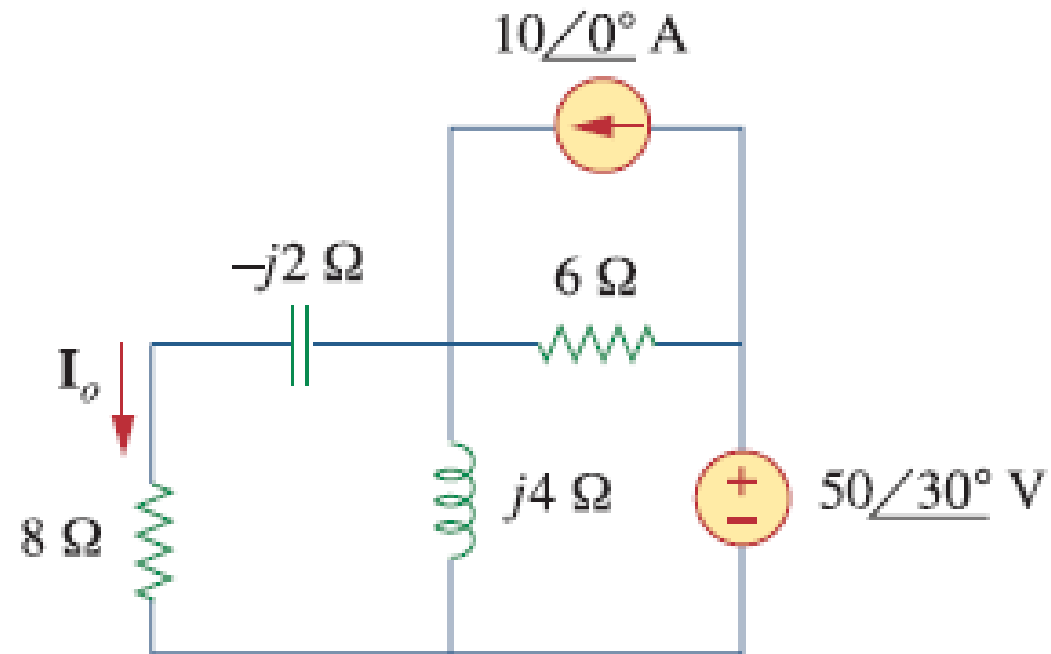
$$\mathbf{I}_o = \mathbf{I}_1 - \mathbf{I}_2 = \frac{\Delta_1 - \Delta_2}{\Delta} = \frac{4}{2(1-j)} = 1+j = 1.4142 \angle 45^\circ$$

Therefore,

$$i_o(t) = \mathbf{1.4142} \cos(2t + 45^\circ) \text{ A}$$

Numerical-3

Find I_o using mesh analysis.



Answer: $5.969\angle 65.45^\circ\text{ A}$.